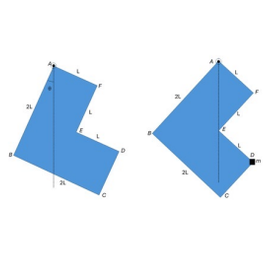
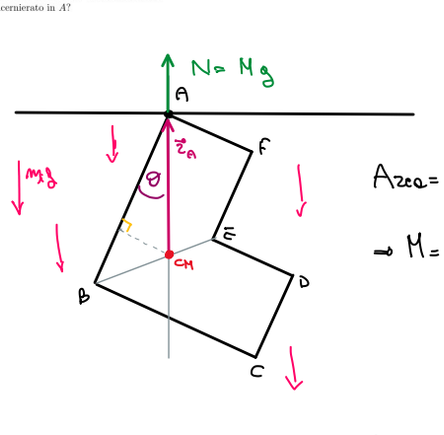
CORPO RIGIDO

mercoledì 4 giugno 2025 11:51 PM

~ DOVE SI TROVA CM ?

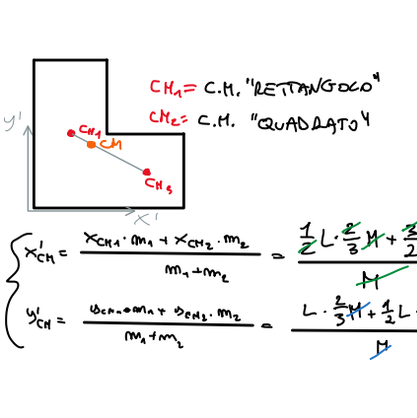
2. Sia data una piastra rigida omogenea di densità superficiale pari 10 kg/m² come in figura, dove L = 20 cm. La piastra è incernierata nel punto A a un asse orizzontale ortogonale alla piastra stessa e intorno al quale può ruotare liberamente senza attrito. Calcolare l'angolo θ che il lato AB della piastra forma con la verticale nella posizione di equilibrio statico (figura sx). Quanto deve valere la massa m di un punto materiale saldato nel vertice D (figura dx) della piastra affinché, sempre all'equilibrio statico, il vertice E cada sulla verticale dell'asse incernierato in A?





Area = $3 \cdot L^2 = 3(0.2\,m)^2 = 0.12\ m^2$

$\Rightarrow M = 10\ Kg\,m^{-2} \cdot 0.12\ m^2 = 1,2\ Kg$



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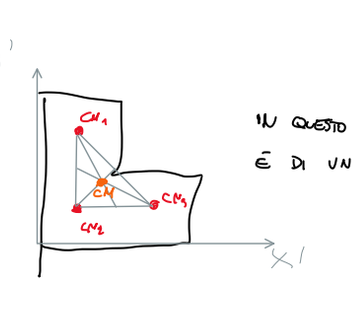
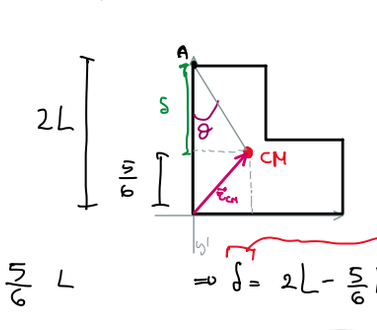


IN QUESTO CASO OGNI CM. È DI UN QUADRATO

$$r_{CM} = \begin{cases} x'_{CM} = \dfrac{x_{CM_1} \cdot m_1 + x_{CM_2} \cdot m_2}{m_1+m_2} = \dfrac{\frac{1}{2}L \cdot \frac{2}{3}M + \frac{3}{2} \cdot L \cdot \frac{1}{3}M}{M} = \frac{5}{6}L \\ y'_{CM} = \dfrac{y_{CM_1} \cdot m_1 + y_{CM_2} \cdot m_2}{m_1+m_2} = \dfrac{L \cdot \frac{2}{3}M + \frac{1}{2}L \cdot \frac{1}{3}M}{M} = \frac{5}{6}L \end{cases}$$

$$r_{CM} = \begin{cases} x'_{CM} = \dfrac{x_{CM_1} \cdot m_1 + x_{CM_2} \cdot m_2 + x_{CM_3} \cdot m_3}{m_1+m_2+m_3} = 2 \cdot \frac{1}{2} \cdot \frac{1}{3}L + \frac{3}{2} \cdot \frac{1}{3}L = \frac{5}{6}L \\ y'_{CM} = \dfrac{y_{CM_1} \cdot m_1 + y_{CM_2} \cdot m_2 + y_{CM_3} \cdot m_3}{m_1+m_2+m_3} = 2\frac{1}{2} \cdot \frac{1}{3}L + \frac{3}{2} \cdot \frac{1}{3}L = \frac{5}{6}L \end{cases}$$



$r_{CM}\cos(45°) = \frac{5}{6}L$

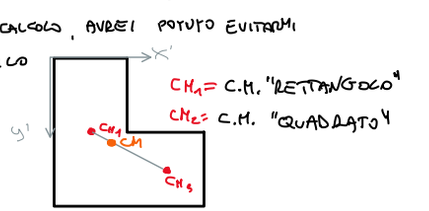
$\Rightarrow r_{CM} = \frac{5}{6}L \cdot \sqrt{2} =$

$\Rightarrow \delta = 2L - \frac{5}{6}L = \frac{7}{6}L$

$\Rightarrow \overline{ACM} = \sqrt{\left(\frac{7}{6}L\right)^2 + \left(\frac{5}{6}L\right)^2}$

$$\Rightarrow \begin{cases} \cancel{\overline{ACM}}\ \text{sen}\,\theta = \frac{5}{6}\cancel{L} \\ \cancel{\overline{ACM}} \cos\theta = \frac{7}{6}\cancel{L} \end{cases} \Rightarrow tg\,\theta = \frac{\frac{5}{6}}{\frac{7}{6}} \Rightarrow \theta = arctg\left(\frac{5}{7}\right) \approx 35,54°$$

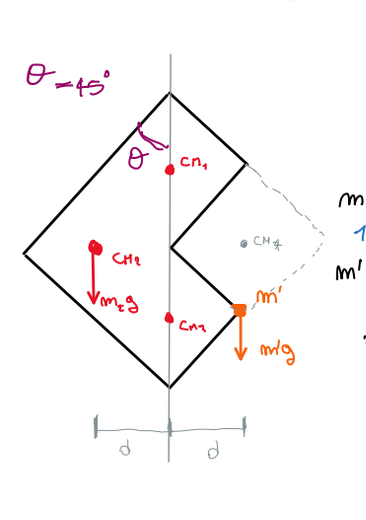
NOTA! IN REALTÀ DURANTE IL CALCOLO DELLA POSIZ. DEL C.M. SAREBBE STATO MEGLIO USARE IL SISTEMA DI RIF. "⌐'", IN QUESTO MODO AVREI AVUTO $x_{CM} = \frac{5}{6}L$ E $y'_{CM} = \frac{7}{6}L$ MA y'_{CM} CORRISPONDEREBBE A "δ" COSÌ DOPO DURANTE IL CALCOLO, AVREI POTUTO EVITARMI DI CALCOLARLO



$$\Rightarrow r_{CM} = \begin{cases} x'_{CM} = \dfrac{x_{CM_1} \cdot m_1 + x_{CM_2} \cdot m_2}{m_1+m_2} = \dfrac{\frac{1}{2}L \cdot \frac{2}{3}M + \frac{3}{2} \cdot L \cdot \frac{1}{3}M}{M} = \frac{5}{6}L \\ y'_{CM} = \dfrac{y_{CM_1} \cdot m_1 + y_{CM_2} \cdot m_2}{m_1+m_2} = \dfrac{L \cdot \frac{2}{3}M + \frac{3}{2}L \cdot \frac{1}{3}M}{M} = \frac{7}{6}L \end{cases}$$

b)

θ = 45°



m' = ?

1° METODO (LOGICO)

m' = (MASSA CHE AVREBBE IL CM 4)

$= m_{CM_1} = m_{CM_2} = m_{CM_3}$

$= \frac{1}{3}M = 0.4\ Kg$

2° METODO (CALCOLO MOMENTI CON POLO CM_1 O CM_3 (QUALSIASI PUNTO SULLA RETTA VERT.))

$\Rightarrow \vec{M} = 0 = \vec{M}_2 + \vec{M}'$

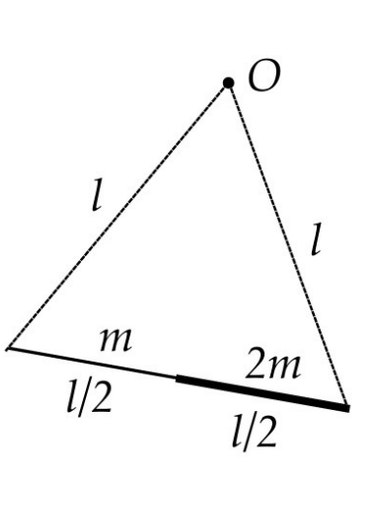
$= m_2 \vec{g} \times \vec{r}_2 + m'\vec{g} \times \vec{r}'$

MA I BRACCI DI $\vec{r}_2$ E $\vec{r}'$ SONO UGUALI E OPPOSTI PERCHÉ LA FIGURA È SIMMETRICA E IL POLO AL CENTRO (BRACCIO = d = r sen φ)

$= m_2 \cancel{g}\cancel{d} - m'\cancel{g}\cancel{d} = 0$

$\Rightarrow m_2 = m' = \frac{1}{3}M = 0.4$

2. Un'asta di lunghezza l composta da due pezzi uniformi di uguale lunghezza $l/2$, ma un pezzo pesa il doppio dell'altro. L'asta è sospesa con cavi di massa trascurabile di lunghezza l fissati a ciascuna estremità e a un chiodo in O, come mostrato in figura. Calcolare l'angolo che l'asta forma con l'orizzontale in condizioni di equilibrio statico.
Supponendo i cavi perfettamente rigidi e assumendo $l = 1$ m, $m = 1$ kg e $g \simeq \pi^2$ m/s², calcolare il periodo delle piccole oscillazioni del sistema rispetto all'asse orizzontale passante per O. Si trascuri ogni forma di attrito.



l = 1 m

m = 1 Kg

$g = \pi^2$

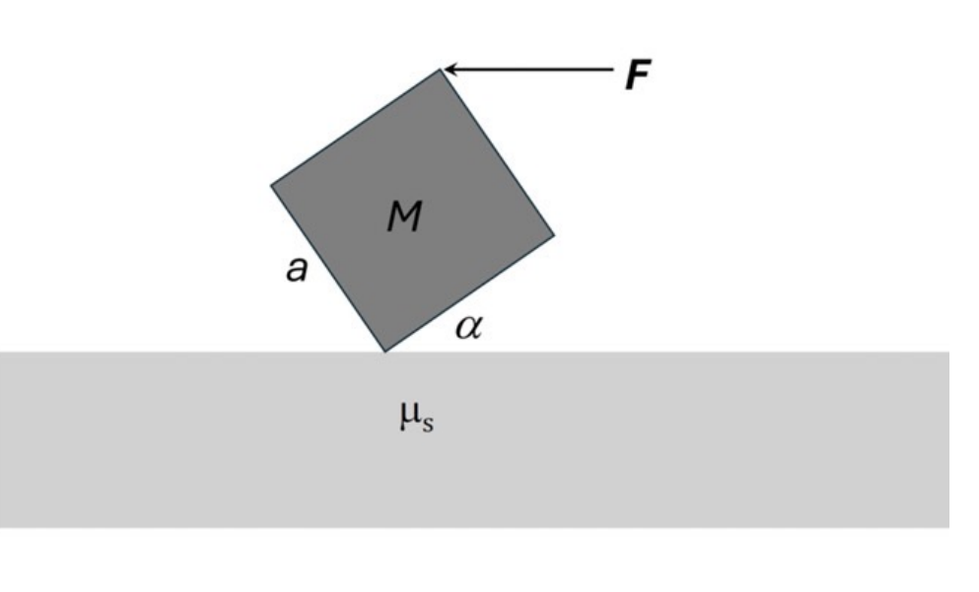
T =

BOH

2. Si consideri un cubo omogeneo avente lo spigolo lungo $a = 50$ cm e massa pari a $M = 2$ kg, inclinato di un angolo $\alpha = 30°$ rispetto allorizzontale (vedi figura). Il corpo mantenuto in equilibrio statico da una forza $\boldsymbol{F}$ applicata orizzontalmente. Sapendo che il piano su cui appoggia scabro con coefficiente di attrito statico μ_s, calcolare:

a) il modulo della forza F in funzione di μ_s;

b) il valore minimo del coefficiente di attrito che consente al cubo di non scivolare.



M = 2 Kg

ℓ = 0.5

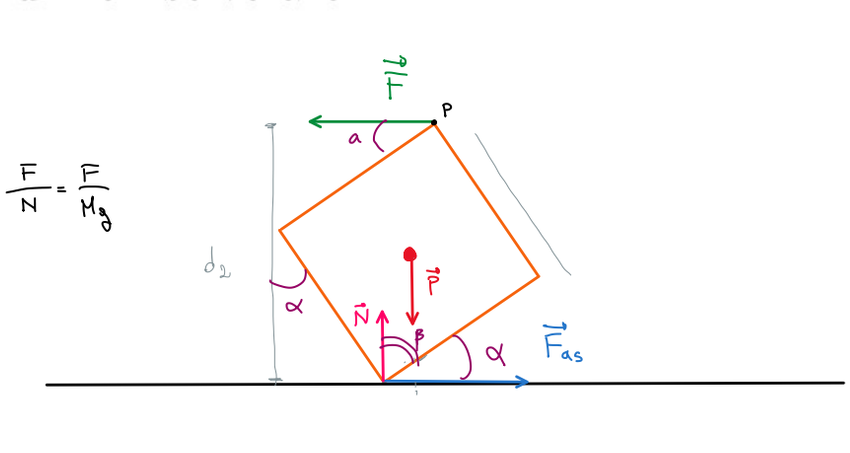
α = 30°

$\vec{R} = \vec{N} + \vec{F}_{as} + \vec{P} + \vec{F}$

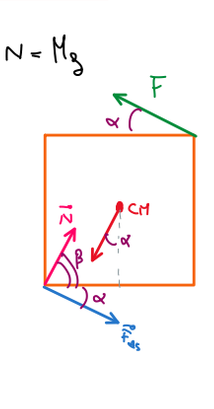
$$\Rightarrow \begin{cases} x: -F + \mu N = 0 \Rightarrow \boxed{F = \mu_s N} \\ y: N - mg = 0 \Rightarrow N = Mg \end{cases} \Rightarrow \mu_s = \frac{F}{N} = \frac{F}{Mg}$$

RISPOSTA 1

POLO: ANGOLO CHE TOCCA TERRA



$\sum \vec{M}_i = 0 \Rightarrow M_F = \vec{F} \cdot \vec{r}_f$ $\boxed{\beta = \frac{\pi}{2} - \alpha}$

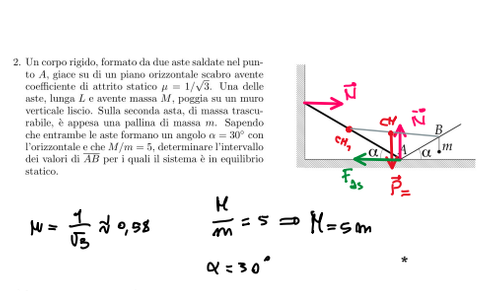


$$\Rightarrow M_F = \begin{cases} M_{F_x} = F\cos\alpha \cdot \ell \\ M_{F_y} = F\,\text{sen}\,\alpha \cdot \ell \end{cases}$$

$$\Rightarrow M_P = \begin{cases} M_{P_x} = -P\,\text{sen}\,\alpha \cdot \frac{\ell}{2} \\ M_{P_y} = -P\cos\alpha \cdot \frac{\ell}{2} \end{cases}$$

$\Rightarrow Mg \cdot \frac{\cancel{\ell}}{2}\cancel{(\text{sen}\,\alpha + \cos\alpha)} = F\cancel{\ell}\cancel{(\text{sen}\,\alpha + \cos\alpha)}$

$\Rightarrow F = \frac{Mg}{2} \Rightarrow \frac{Mg}{F} = 2 \Rightarrow \frac{F}{Mg} \simeq 0.5 = \mu$



$\mu = \frac{1}{\sqrt{3}} \simeq 0,58$ $\frac{M}{m} = 5 \Rightarrow M = 5m$

α = 30°

M = massa asta 1

L = LUNGHEZZA ASTA 1

0 = MASSA ASTA 2

$I \ni \overline{AB}$ t.c. $\left\{ \begin{matrix} \vec{R} = 0 \\ \vec{M} = 0 \end{matrix} \right\}$

$Max\{\overline{AB}\} = \frac{L}{2}$

MASSA TOT = m + 5m = 6m $\Rightarrow \vec{P} = 6m\vec{g} \Rightarrow P = (58,86\,m)$ N

$\vec{R} = \vec{N}_x + \vec{N}_y + \vec{P} + \vec{F}_{as}$

$$\Rightarrow \begin{cases} x: N_x - \mu N_y = 0 \Rightarrow N_x = \mu N_y = 6mg\mu \\ y: -6mg + N_y = 0 \Rightarrow N_y = \frac{N_x}{\mu} = 6mg \end{cases}$$

POLO: PUNTO DI APPOGGIO SULLE x

$\vec{M} = \vec{M}_{N_x} + \vec{M}_{CM_1} + \vec{M}_m = 0$

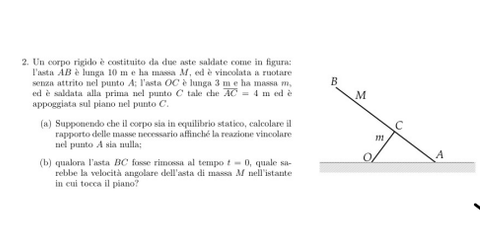
- $M_{N_x} = -N_x \cdot L\,\text{sen}\,\alpha = -6mg\mu \cdot L\,\text{sen}\,\alpha$
- $M_{CM_1} = 5mg \cdot \frac{L}{2}\cos\alpha$
- $M_m = -mg \cdot \overline{AB}\cos\alpha$

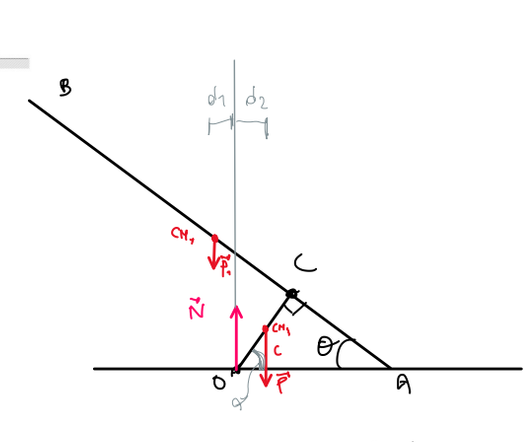
$\Rightarrow mg\cos\alpha\left(-6L\mu\,tg\,\alpha + 5\frac{L}{2} - \overline{AB}\right) = 0$

$\Rightarrow \overline{AB} = \left(\frac{5}{2} - 6\mu\,tg\,\alpha\right)L$

I DUE CASI LIMITE SONO: $N_x = F_{as} \Rightarrow \overline{AB} = \left(\frac{5}{2} - 6\mu\,tg\,\alpha\right)L = \frac{1}{2}L$

$N_x = 0 \Rightarrow \overline{AB} = \left(\frac{5}{2} - \underbrace{6\mu\,tg\,\alpha}_{0 \text{ PERCHÉ } N_x = 0 = 6mg\mu}\right)L = \frac{5}{2}L$





AB = 10 m massa AB = M

OC = 3 m massa OC = m

AC = 4 m

a) BC = 6 m

$\vec{R} = 0 \Rightarrow N + m_{tot}\vec{g}$

$\Rightarrow (m + M)g = N$

$\Rightarrow M$ POLO: O

$\theta = arctg\left(\frac{3}{4}\right) \simeq 37°$

$AO = \sqrt{AC^2 + OC^2} = 5$ m

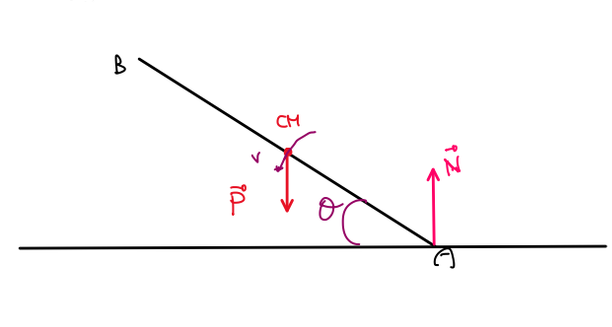
$\Rightarrow \sum \vec{M} = 0 \Rightarrow M_{CM_2} = -mg\,\frac{OC}{2} \cdot \text{sen}\,\theta$

$M_{CM_1} = Mg\,\frac{AB}{2}\cos\theta - AO$

$\Rightarrow M\cancel{g}\,\frac{AB}{2}\cos\theta - AO = m\cancel{g}\,\frac{OC}{2} \cdot \text{sen}\,\theta$

$\Rightarrow \frac{M}{m} = \frac{OC}{AB} \cdot tg\,\theta + \frac{2AO}{AB\cos\theta} \approx 1,48$

b



$K_f = \frac{1}{2} m\,\omega^2 R^2$

ω = ?

$\vec{R} = \vec{P} + \vec{N} = 0 \Rightarrow N = mg$

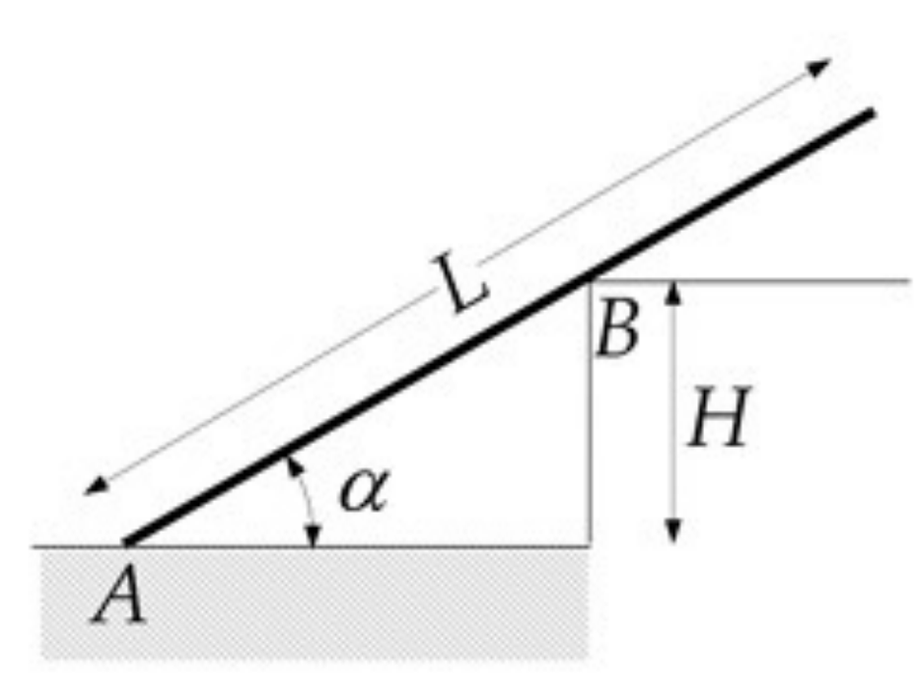
$$\begin{cases} T: \cancel{m}g\,\text{sen}\,\theta = \cancel{m}\frac{dv(\theta)}{dt} \Rightarrow g\int_0^T \cos\theta\,dt = v(\theta) \Rightarrow \int_0^T \cos\theta\,dt = \frac{v(\theta)}{g} \\ C: \cancel{m}g\cos\theta = \cancel{m}\frac{v^2(\theta)}{R} \Rightarrow \omega^2(\theta) = \frac{g\cos\theta}{R} \Rightarrow \omega(\theta) = \sqrt{\frac{g\cos\theta}{R}} \Rightarrow \text{SE } \theta = 0 \Rightarrow \omega(0) = \sqrt{\frac{g}{R}} \simeq 1,4\ \text{rad/s} \end{cases}$$

$M_P = mg\,\frac{AB}{2}\cos\theta$

$\Rightarrow \frac{dM_P}{dt} = L = mg \cdot \frac{AB}{2}\int_0^T \cos\theta\,dt$

$= mg \cdot \frac{AB}{2} \cdot \frac{v(\theta)}{g}$

Un'asta omogenea di lunga 100 cm è appoggiata a un pavimento scabro e allo spigolo perfettamente liscio di uno scalino alto 30 cm, come mostrato in figura. Si osserva che l'asta scivola non appena l'angolo α diventa minore di 30°. Determinare il coefficiente di attrito statico nel punto A.



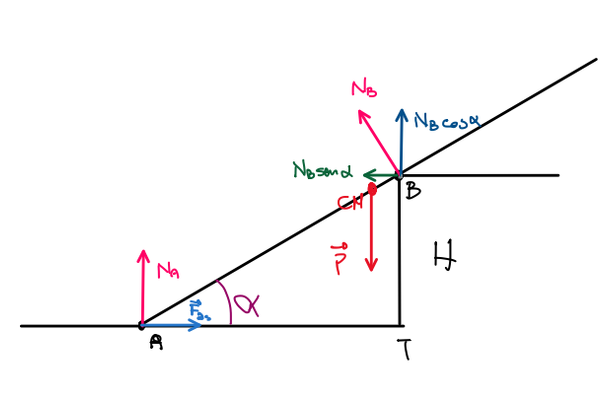
L = 1 m

H = 0.3 m

$\alpha_{min} = 30°$

$\mu_s = ?$

$\Rightarrow \vec{R} = 0 = \vec{N}_A + \vec{N}_B + \vec{P} + \vec{F}_{as}$



$$\begin{cases} \mu_s N_A = N_B\,\text{sen}\,\alpha \Rightarrow \mu_s = \frac{N_B\,\text{sen}\,\alpha}{N_A} \\ N_A + N_B\cos\alpha = mg \end{cases}$$

$\Rightarrow N_B = \frac{mg - N_A}{\cos\alpha} \Rightarrow \mu_s = \frac{mg - N_A}{\cos\alpha} \cdot \frac{\text{sen}\,\alpha}{N_A}$

$= \frac{mg - N_A}{N_A} \cdot tg\,\alpha$

$\sum \vec{M}_i = 0$

POLO B

$M_{N_A} = N_A \cdot H \cdot cotg\,\alpha$

$M_{F_{as}} = \mu_s N_A \cdot H$

$M_{CM} = mg\left(H \cdot cotg\,\alpha - \frac{1}{2}L\cos\alpha\right)$

$\Rightarrow mg\left(H \cdot cotg\,\alpha - \frac{1}{2}L\cos\alpha\right) + \mu_s N_A \cdot H - N_A \cdot H\ cotg(\alpha) = 0$

$\Rightarrow \frac{mg}{N_A H}\left(H \cdot cotg\,\alpha - \frac{1}{2}L\cos\alpha\right) + \mu_s - cotg(\alpha) = 0$

$\Rightarrow \mu_s = arctg(\alpha) - \frac{mg}{N_A}\left(cotg(\alpha) - \frac{L}{2H}\cos\alpha\right)$

$\Rightarrow \left(\frac{mg}{N_A} - 1\right)tg\,\alpha + \frac{mg}{N_A}\left(cotg(\alpha) - \frac{L}{2H}\cos\alpha\right) - arctg(\alpha) = 0$

$\Rightarrow \underset{\frac{\sqrt{3}}{3}}{tg\,\alpha}\,\frac{mg}{N_A} + \frac{mg}{N_A}\underbrace{\left(cotg(\alpha) - \frac{L}{2H}\cos\alpha\right)}_{\frac{\sqrt{3}}{6}} - \overbrace{cotg(\alpha) - tg(\alpha)}^{0} = 0$ $\frac{4\sqrt{3}}{3}$

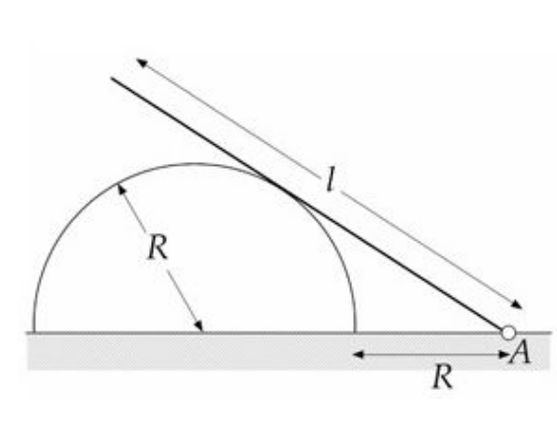
$\Rightarrow \frac{mg}{N_A}\left(tg\,\alpha + cotg\,\alpha - \frac{L}{2H}\cos\alpha\right) = cotg\,\alpha + tg\,\alpha$

$\Rightarrow \frac{mg}{N_A} = \frac{cotg\,\alpha + tg\,\alpha}{tg\,\alpha + cotg\,\alpha - \frac{L}{2H}\cos\alpha}$

$\Rightarrow \frac{mg}{N_A} = \frac{8}{3}$

$\Rightarrow \mu_s = \left(\frac{8}{3} - 1\right)tg\,\alpha \simeq 0.96$

2. Un'asta omogenea di lunghezza $l = 30$ cm e peso di 4 N è incernierata per l'estremo A a un pavimento orizzontale e appoggiata su una superficie semicilindrica (raggio $R = 10$ cm) perfettamente liscia saldata al pavimento, come mostrato in figura. Sapendo che la distanza tra il centro del semicilindro e la cerniera è pari a 20 cm, calcolare le reazioni vincolari agenti sull'asta.

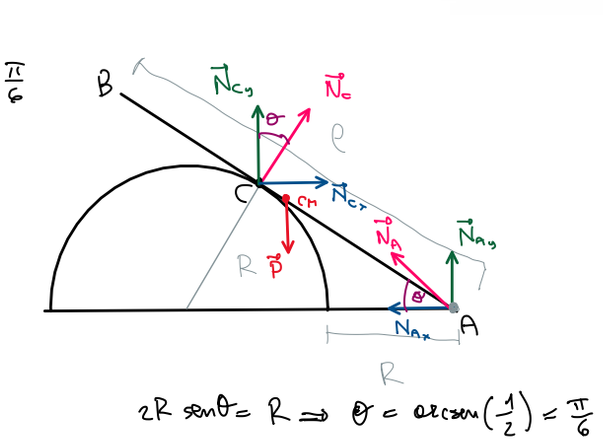


ℓ = 0.3 m $\alpha = 30° = \frac{\pi}{6}$

mg = 4 N ⇒ m ≃ 0.4 Kg

R = 0.1 m

$\Rightarrow \overline{AC} = d = \sqrt{(2R)^2 - (R)^2} = 0.17$ m



$\Rightarrow R = 0 = \vec{N}_A + \vec{N}_C + \vec{P}$ $2R\,\text{sen}\,\theta = R \Rightarrow \theta = arcsen\left(\frac{1}{2}\right) = \frac{\pi}{6}$

$$\Rightarrow \begin{cases} x: N_C\,\text{sen}\,\theta = N_{A_x} \\ y: N_C\cos\theta + N_{A_y} = mg \end{cases}$$

$\sum \vec{M}_i = 0 \Rightarrow$ POLO: A

$\vec{M}_P = mg \cdot \frac{\ell}{2}\cos\theta = \frac{0.3}{2} \cdot \frac{\sqrt{3}}{2}\,mg = \frac{3\sqrt{3}}{40}\,mg$

$M_{\vec{N}_{C_x}} = -N_C\,\text{sen}\,\theta \cdot R\cos\theta = -N_C \cdot \frac{1}{2} \cdot 0.1 \cdot \frac{\sqrt{3}}{2} = -\frac{\sqrt{3}}{40}N_C$

$M_{\vec{N}_{C_y}} = -N_C\cos\theta \cdot (2R - R\,\text{sen}\,\theta) = -N_C \cdot \frac{\sqrt{3}}{2} \cdot 0.1\left(2 - \frac{1}{2}\right) = \frac{3\sqrt{3}}{40}N_C$

$\Rightarrow \frac{3\sqrt{3}}{40}\,mg = N_C\left(\frac{\sqrt{3}}{40} + \frac{3\sqrt{3}}{40}\right)$

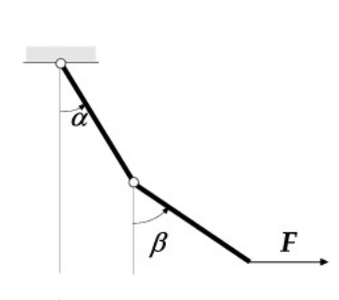
$\Rightarrow N_C = \frac{3\sqrt{3}\,mg}{40\left(\frac{\sqrt{3}}{40} + \frac{3\sqrt{3}}{40}\right)} = \frac{3\cancel{\sqrt{3}}\,4N}{\cancel{\sqrt{3}} \cdot 4} = 3N$

$\Rightarrow N_{A_x} = N_C\,\text{sen}\,\theta = 3N \cdot \frac{1}{2} \simeq 1,5$ N

$\Rightarrow N_{A_y} = mg - N_C\cos\theta = 4N - 3N \cdot \frac{\sqrt{3}}{2} \simeq 1,4$ N

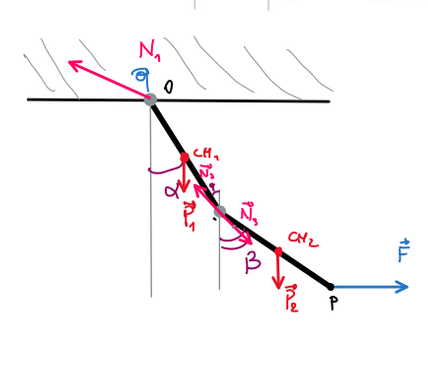
$\Rightarrow N_A = \sqrt{N_{A_x}^2 + N_{A_y}^2} \simeq 2$ N

Due aste rigide omogenee identiche, del peso di 10 N, sono incernierate come in figura e mantenute in equilibrio statico da una forza orizzontale $\boldsymbol{F}$ di 10 N. Calcolare, in queste condizioni, il valore degli angoli α e β e le reazioni vincolari nelle due cerniere. Si trascuri ogni forma di attrito.



$P_1 = P_2 = F = 10$ N

$\vec{R} = \vec{P}_1 + \vec{P}_2 + \vec{N}_1 + \overbrace{\vec{N}_2 + \vec{N}_3}^{0} + F = 0$



$$\begin{cases} x: N_{1x} = F = 10\,N \\ y: N_{1y} = 2mg = 20\,N \end{cases}$$

$\sum \vec{M}_i = 0 \Rightarrow$ POLO: CERNIERA INTERNA

ASTA 1

$\Rightarrow M_{N_{1y}} = -N_{1y} \cdot \ell\,\text{sen}\,\alpha$

$\Rightarrow M_{N_{1x}} = N_{1x} \cdot \ell\cos\alpha$

$\Rightarrow M_{P_1} = mg\ell \cdot \frac{1}{2}\text{sen}\,\alpha$

ASTA 2

$\Rightarrow M_{P_2} = -mg\ell \cdot \frac{1}{2}\text{sen}\,\beta$

$\Rightarrow M_F = F\ell\cos\beta$

ASTA 1:

$$\begin{cases} x: N_{1x} = N_{3x} = 10\,N \\ y: N_{1y} = mg + N_{3y} \Rightarrow N_{3y} = 10\,N \end{cases}$$

ASTA 2:

$$\begin{cases} x: F = N_{2x} = 10\,N \\ y: N_{2y} = mg = 10\,N \end{cases}$$

$\Rightarrow -2\,\text{sen}\,\alpha + \cos\alpha + \frac{1}{2}\text{sen}\,\alpha = 0$

$\Rightarrow \frac{3}{2}\text{sen}\,\alpha = \cos\alpha \Rightarrow tg\,\alpha = \frac{2}{3}$

$\Rightarrow \alpha \simeq 33,69°$

$\Rightarrow \cos\beta = \frac{1}{2}\text{sen}\,\beta$

$\Rightarrow tg\,\beta = 2$

$\Rightarrow \beta = 63,44°$

$\Rightarrow$ 3 REAZIONI VINCOLARI:

$\vec{N}_1 = (-10N, 20N)$

$\vec{N}_2 = (-10N, 10N)$ ⎫ UGUALI E OPPOSTE

$\vec{N}_3 = (10N, -10N)$ ⎭

ALTRO MODO CON ENERGIA POTENZIALE

$U = -mg(h_1 + h_2) - F \cdot x = -m\,g\ell\left(\frac{1}{2}\cos\alpha + \cos\alpha + \frac{1}{2}\cos\beta\right) - F\ell(\text{sen}\,\alpha + \text{sen}\,\beta)$

$U_{(F)} = -F \cdot d = -F \cdot \ell(\text{sen}\,\alpha + \text{sen}\,\beta)$

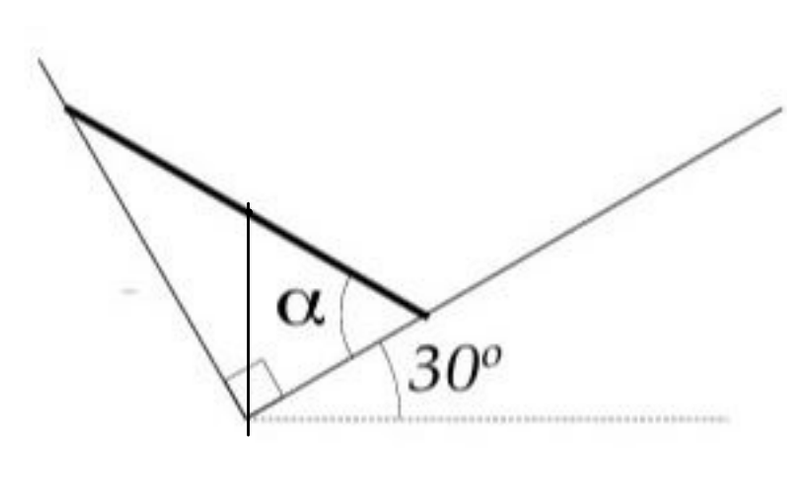
$= -mg\ell\left(\frac{3}{2}\cos\alpha + \frac{1}{2}\cos\beta\right) - \overset{mg\ell}{\overset{\shortparallel}{F\ell}}(\text{sen}\,\alpha + \text{sen}\,\beta)$

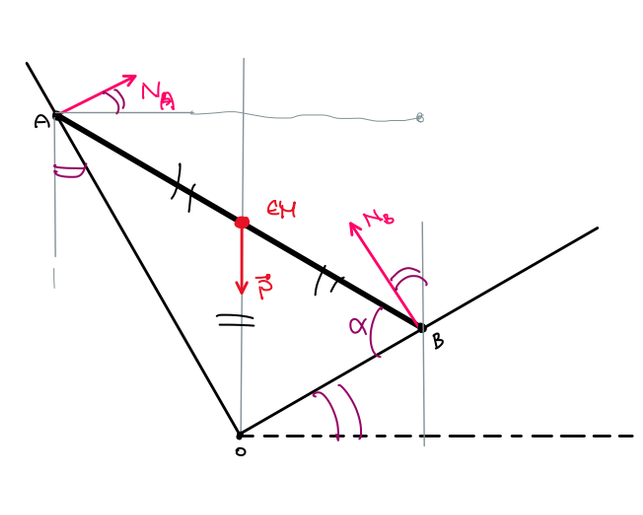
$= -mg\ell\left(\frac{3}{2}\cos\alpha + \frac{1}{2}\cos\beta + \text{sen}\,\alpha + \text{sen}\,\beta\right)$

$\Rightarrow \frac{\partial}{\partial\alpha}\left(-mg(h_1 + h_2) - Fx\right) = -mg\ell\,\frac{\partial}{\partial\alpha}\left(\frac{3}{2}\cos\alpha + \text{sen}\,\alpha\right) = 0 \Rightarrow mg\ell\left(\frac{3}{2}\text{sen}\,\alpha - \cos\alpha\right) = 0 \Rightarrow tg\,\alpha = \frac{2}{3} \Rightarrow \alpha \simeq 33,69°$

$\frac{\partial}{\partial\beta}\left(-mg(h_1 + h_2) - Fx\right) = -mg\ell\,\frac{\partial}{\partial\beta}\left(\frac{1}{2}\cos\beta + \text{sen}\,\beta\right) = 0 \Rightarrow mg\ell\left(\frac{1}{2}\text{sen}\,\beta - \cos\beta\right) = 0 \Rightarrow tg\,\beta = 2 \Rightarrow \beta \simeq 63,44°$

2. Un'asta omogenea di massa M è in equilibrio statico appoggiata a due piani perfettamente lisci mutuamente ortgonali, come mostrato in figura. Calcolare l'angolo di equilibrio α e le reazioni vincolari.



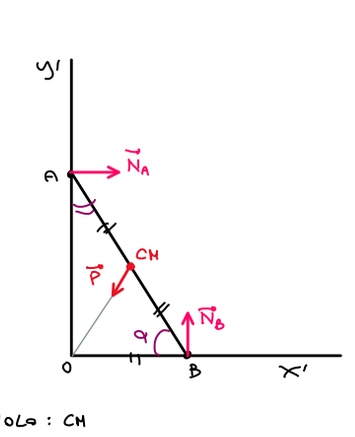


$\Rightarrow \vec{R} = \vec{N}_A + \vec{N}_B + \vec{P} = 0$

$$\Rightarrow \begin{cases} x: N_B\,\text{sen}(30°) = N_A\cos(30°) \Rightarrow N_B = \sqrt{3}N_A \\ y: N_A\,\text{sen}(30°) + N_B\cos(30°) = Mg \Rightarrow \frac{1}{2}N_A + \frac{3}{2}N_A = Mg \Rightarrow 4N_A = 2Mg \Rightarrow N_A = \frac{1}{2}Mg \end{cases}$$

$\Rightarrow N_B = \frac{\sqrt{3}}{2}Mg$

SISTEMA x', y'

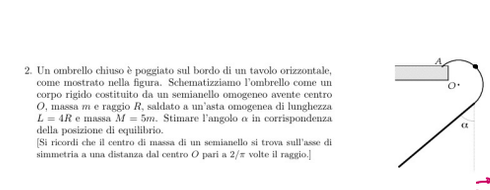


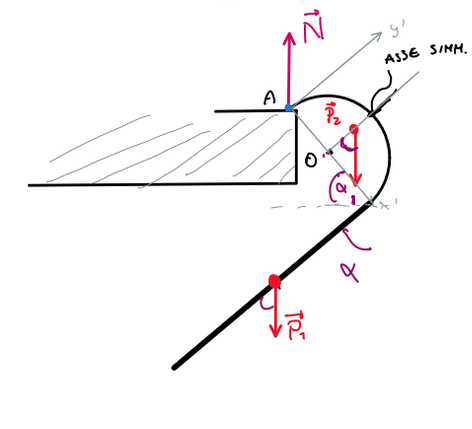
$\sum \vec{N}_i = 0 \Rightarrow$ POLO: CM

$\Rightarrow M_{N_A} = -N_A \cdot \frac{\ell}{2}\text{sen}\,\alpha$

$M_{N_B} = N_B \cdot \frac{\ell}{2}\cos\alpha$

$\Rightarrow \cancel{N_A}\,\text{sen}\,\alpha = \sqrt{3}\cancel{N_A}\cos\alpha \Rightarrow tg\,\alpha = \sqrt{3} \Rightarrow \alpha = 60°$





ANELLO: O, m, R

ASTA: M = 5m, L = 4R

$\Rightarrow \vec{R} = \vec{P}_1 + \vec{P}_2 + \vec{N} = 0$

$\Rightarrow N = 5mg + mg = 6mg$

POLO: A

$\Rightarrow M_{\vec{P}_{2x}} = -mg \cdot \cos\alpha \cdot \frac{2}{\pi}R$

$M_{\vec{P}_{2y}} = -mg \cdot \text{sen}\,\alpha \cdot R$

$M_{\vec{P}_{1x}} = 5mg\,\text{sen}\,\alpha \cdot 2R$

$M_{\vec{P}_{1y}} = -5mg\cos\alpha \cdot 2R$

$\Rightarrow \frac{2}{\pi}\cos\alpha + \text{sen}\,\alpha + 10\cos\alpha - 10\,\text{sen}\,\alpha = 0$

$\Rightarrow \cos\alpha\left(\frac{2}{\pi} + 10\right) = 9\,\text{sen}\,\alpha$

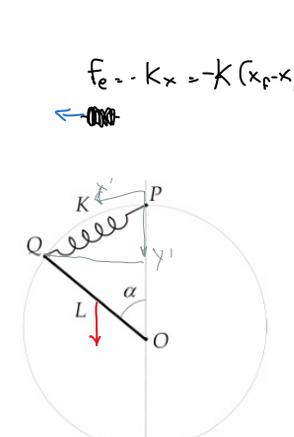
$\Rightarrow tg\,\alpha = \frac{\left(\frac{2}{\pi} + 10\right)}{9} \Rightarrow \alpha \simeq 50°$

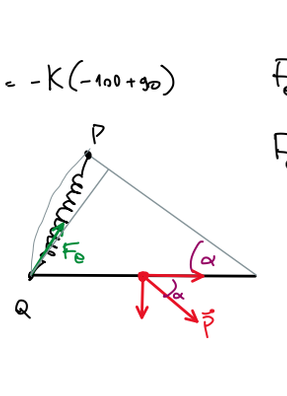
$F_e = Kx = -K(x_f - x_i) = -K(-100 + 90)$

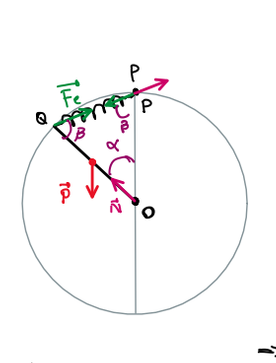
$F_{e_x} =$

$F_{e_y} =$

(b) Un'asta omogenea di massa M e lunghezza L può ruotare senz'attrito attorno a un asse orizzontale passante per l'estremo fisso O. L'estremo libero Q è agganciato al punto (fisso) P tramite una molla avente costante elastica K e lunghezza a riposo nullo. Calcolare i valori dell'angolo α corrispondenti alle posizioni di equilibrio statico. [Fac.: discutere la stabilità dell'equilibrio]







$\int_x^0 Kx\,dx$

OQ = L

OP = L

PQ = ?

$OQ_x = L\,\text{sen}\,\alpha \Rightarrow \boxed{PQ_x = L\,\text{sen}\,\alpha}$

$OQ_y = L\cos\alpha \Rightarrow \boxed{PQ_y = L(1 - \cos\alpha)}$

$\Rightarrow F_{ex} = KL\,\text{sen}\,\alpha$

$F_{ey} = KL(1 - \cos\alpha)$

$F_{ex} = KL(\text{sen}\,\alpha)$

$F_{ey} = KL(1 - \cos\alpha)$

⇑

$|\vec{F}_e| = \sqrt{K^2L^2(\text{sen}\,\alpha)^2 + K^2L^2(1 - \cos\alpha)^2}$

$= KL\sqrt{\text{sen}^2\alpha + \cos^2\alpha - 2\cos\alpha + 1}$

$= KL\sqrt{2 - 2\cos\alpha}$

$= \sqrt{2}KL\sqrt{1 - \cos\alpha}$

$x = \sqrt{2}L\sqrt{1 - \cos\alpha}$

$\Rightarrow U_{F_e} = \int_x^0 Kx\,dx = -\frac{1}{2}Kx^2$

$U_P = -mg\int_h^0 dx = -$

$\Rightarrow U_{F_e} = \frac{1}{2}Kx^2 = \frac{1}{2}K \cdot (2L^2(1 - \cos\alpha)) = KL^2(1 - \cos\alpha)$

$\Rightarrow \vec{R} = \vec{F}_e + \vec{P} + \vec{N} = 0$

$$\Rightarrow \begin{cases} x: KL\,\text{sen}\,\alpha = N\,\text{sen}\,\alpha \Rightarrow N = KL \\ y: KL(1 - \cos\alpha) + KL\cos\alpha = Mg \Rightarrow Mg = KL \end{cases}$$

$\sum \vec{M}_i = 0$ POLO: O

$M_P = Mg \cdot \frac{\ell}{2}\text{sen}\,\alpha$

$M_{F_{e_x}} = -KL\,\text{sen}\,\alpha \cdot \ell\cos\alpha$

$M_{F_{e_y}} = -KL(1 - \cos\alpha) \cdot \ell\,\text{sen}\,\alpha$

$\Rightarrow \frac{1}{2}\text{sen}\,\alpha$ ~~$-\text{sen}\,\alpha\cos\alpha - \text{sen}\,\alpha + \text{sen}\,\alpha\cos\alpha$~~ $= 0$

$\Rightarrow$ 2 SOLUZIONI $\begin{cases} \alpha = 0 \\ \alpha = \pi \end{cases}$

METODO CON ENERGIA POTENZIALE

$U_{TOT} = mgh + \frac{1}{2}MgL\cos\alpha +$

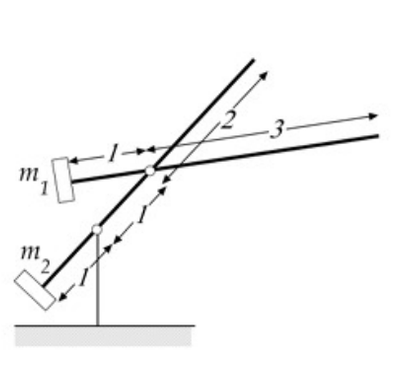
$\Rightarrow U_{TOT} = \frac{1}{2}KL^2(\cos\alpha) + mg \cdot L(1 - \cos\alpha)$

$\frac{d}{d\alpha} = KL^2 \cdot \frac{d}{d\alpha}\left(\frac{1}{2}\cos\alpha + 1 - \cos\alpha\right) = 0$

$\Rightarrow \frac{1}{2}\text{sen}\,\alpha + \text{sen}\,\alpha = 0 \Rightarrow \alpha = \begin{cases} 0 \\ \pi \end{cases}$

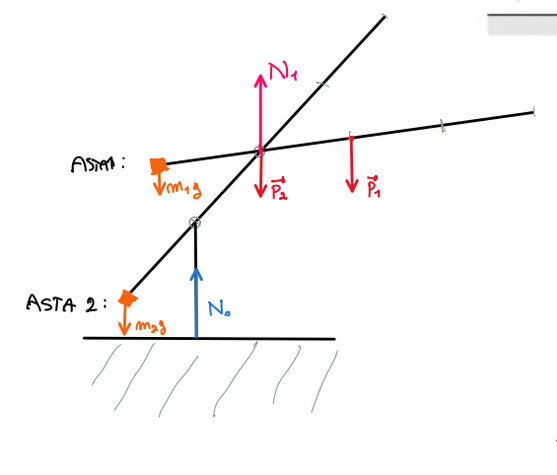
$\Rightarrow \frac{d}{d\alpha}\left(\frac{3}{2}\text{sen}\,\alpha\right) = \frac{3}{2}\cos\alpha \begin{cases} > 0 \text{ SE } \alpha = 0 \\ < 0 \text{ SE } \alpha = \pi \end{cases}$

Due aste rigide omogenee identiche, del peso di 10 N, sono incernierate come in figura. Calcolare i valori delle masse (considerate puntiformi) m_1 ed m_2 che devono essere saldate alle due aste in modo che il sistema complessivo sia in equilibrio statico per arbitrarie angolazioni delle aste. Si trascuri ogni forma di attrito.



$P_1 = P_2 = 10$ N

$\Rightarrow m_{ASTA} = 1$ Kg



ASTA 1: $\sum \vec{M}_i = 0 \Rightarrow M_{P_1} = -m_{ASTA}\,g \cdot \frac{L}{4}\cos\alpha$

$M_{m_1} = m_1 g \cdot \frac{L}{4}\cos\alpha$

$\Rightarrow m_1 = m_{ASTA} \simeq 1$ Kg

ASTA 2: $\sum \vec{M}_i = 0$

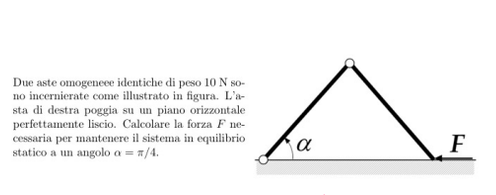
$\Rightarrow M_{P_2} = -m_{ASTA}\,g \cdot b$

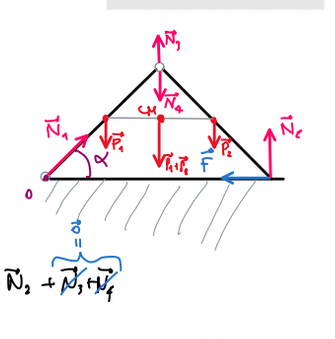
$M_{(P_1 + m_1 g)} = -(m_{ASTA}\,g + m_1 g) \cdot b$

$M_{(m_2 g)} = m_2 g \cdot b$

$\Rightarrow -m_{ASTA} - m_{ASTA} - m_1 + m_2 = 0$

$\Rightarrow m_2 = 3m_1 \simeq 3$ Kg





$\Rightarrow \vec{R} = \vec{N}_1 + \vec{P}_1 + \vec{P}_2 + \vec{N}_3 + \overbrace{\vec{N}_1 + \vec{N}_2}^{0}$

$$\Rightarrow \begin{cases} x: \frac{1}{\sqrt{2}}N_1 = F \\ y: \frac{1}{\sqrt{2}}N_1 + N_2 = 2mg \end{cases}$$

$\Rightarrow N_2 = 2mg - F$

$\sum \vec{M}_i = 0$ POLO: O

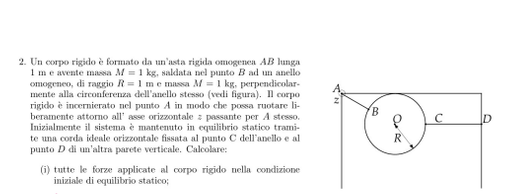
$\Rightarrow M_{\vec{P}_{1,2}} = 2mg \cdot \frac{1}{\sqrt{2}}\ell = \frac{2}{\sqrt{2}}mg\ell = \sqrt{2}\,mg\ell$

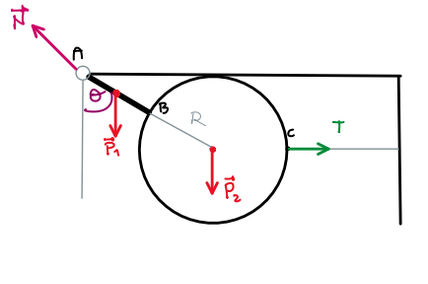
$M_{\vec{N}_2} = N_2 \cdot \frac{2}{\sqrt{2}}\ell = \frac{2}{\sqrt{2}}(2mg - F)\ell = (2\sqrt{2}\,mg - \sqrt{2}F)\ell$

$M_F = F \cdot 0 = 0$

$\Rightarrow 2\sqrt{2}\,mg - \sqrt{2}\,mg - \sqrt{2}F = 0$

$\Rightarrow F = 2mg - mg = mg$





$M_1 = M_2 = 1$ Kg

ℓ = R = 1 m

$\vec{R} = 0 = \vec{P}_1 + \vec{P}_2 + \vec{N} + \vec{T}$

$$\begin{cases} x: N_x = T \\ y: N_y = 2Mg \end{cases}$$

$\sum \vec{M}_i = 0$ POLO: CENTRO ANELLO

$M_{\vec{P}_1} = Mg \cdot \frac{3}{4} \cdot 2R\,\text{sen}\,\theta$

$M_{\vec{N}_x} = N_x \cdot 2R\cos\theta = T \cdot 2R\cos\theta$

$M_{\vec{N}_y} = -N_y \cdot 2R\,\text{sen}\,\theta = -2Mg \cdot 2R\,\text{sen}\,\theta$

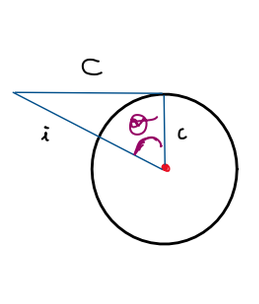
$\Rightarrow Mg \cdot \frac{3}{4}\text{sen}\,\theta - 2Mg\,\text{sen}\,\theta + T\cos\theta = 0$

$\Rightarrow Mg\,\text{sen}\,\theta\left(\frac{3}{4} - 2\right) + T\cos\theta = 0$

$\Rightarrow T\cos\theta = \frac{5}{4}Mg\,\text{sen}\,\theta \Rightarrow T = \frac{5}{4}tg\,\theta$ [N]

QUANTO VALE θ? SO CHE $i = 2R$

$c = R$



MA $c = i\cos\theta \Rightarrow \cos\theta = \frac{c}{i}$

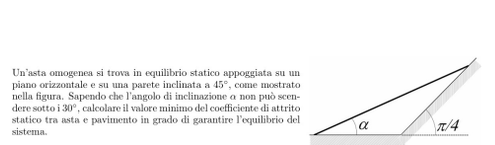
$\Rightarrow \theta = arccos\left(\frac{1}{2}\right) = \frac{\pi}{3}$

$\Rightarrow T = \frac{5}{4} \cdot tg\left(\frac{\pi}{3}\right) \simeq 2.16$ N

$\Rightarrow N_x = 2,16$ N

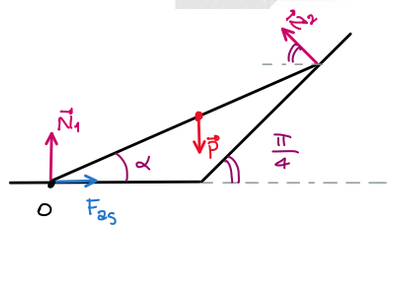
$\Rightarrow N_y = 2$ N

$\Rightarrow N = \sqrt{N_x^2 + N_y^2} \simeq 3$ N



CASO LIMITE: α = 30°

$\vec{R} = \vec{N}_1 + \vec{N}_2 + \vec{P} + \vec{F}_{as} = 0$



$\vec{R} = 0$

$$\Rightarrow \begin{cases} x: \mu N_1 = \frac{1}{\sqrt{2}}N_2 \\ y: N_1 + \frac{1}{\sqrt{2}}N_2 = mg \Rightarrow N_1 + \mu N_1 = mg \Rightarrow mg = N_1(1 + \mu) \end{cases}$$

$\sum \vec{M}_i = 0$; POLO: O

$\Rightarrow M_{\vec{P}} = -mg \cdot \frac{\ell}{2}\cos\alpha = \ell\left(-\frac{\sqrt{3}}{4}mg\right)$

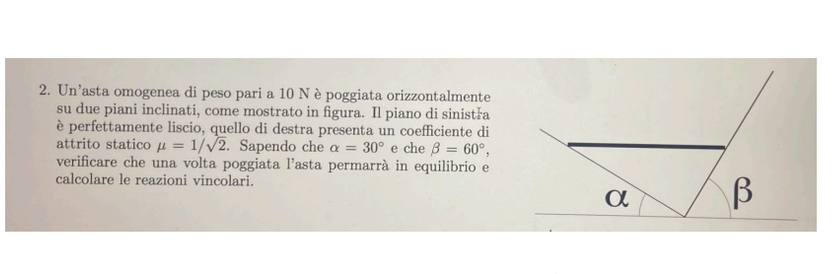
$\Rightarrow M_{\vec{N}_{2x}} = \frac{1}{\sqrt{2}}N_2 \cdot \ell\,\text{sen}\,\alpha = \ell\left(\frac{1}{2} \cdot \frac{1}{\sqrt{2}}N_2\right)$

$M_{\vec{N}_{2y}} = \frac{1}{\sqrt{2}}N_2 \cdot \ell\cos\alpha = \ell\left(\frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}}N_2\right)$

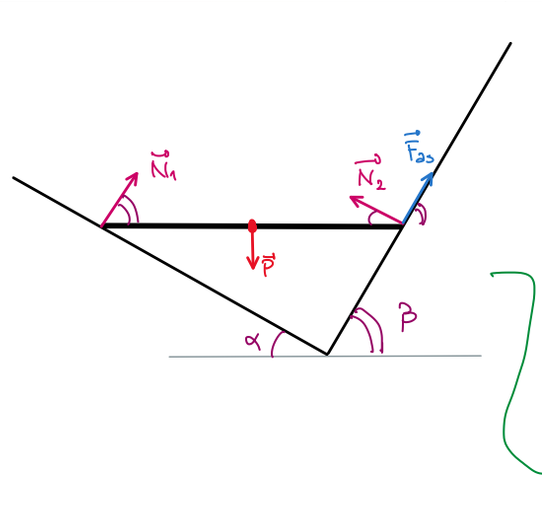
$\Rightarrow \frac{1}{2} \cdot \mu\cancel{N_1} + \frac{\sqrt{3}}{2}\mu\cancel{N_1} = \frac{\sqrt{3}}{4}\cancel{N_1}(1 + \mu)$

$\Rightarrow \mu + \sqrt{3}\mu = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}\mu$

$\Rightarrow \mu\left(1 + \sqrt{3} - \frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{2} \Rightarrow \mu = \frac{\sqrt{3}}{\cancel{2}} \cdot \frac{\cancel{2}}{2 + \sqrt{3}} \simeq 0.46$

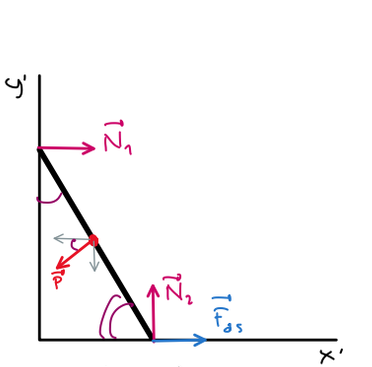


2. Un'asta omogenea di peso pari a 10 N è poggiata orizzontalmente su due piani inclinati, come mostrato in figura. Il piano di sinistra è perfettamente liscio, quello di destra presenta un coefficiente di attrito statico $\mu = 1/\sqrt{2}$. Sapendo che $\alpha = 30°$ e che $\beta = 60°$, verificare che una volta poggiata l'asta permarrà in equilibrio e calcolare le reazioni vincolari.



α = 30°

β = 60°



$\vec{R} = \vec{N}_1 + \vec{N}_2 + \vec{F}_{as} + \vec{P} = 0$

$$\Rightarrow \begin{cases} x': N_1 + F_a = mg\cos(30°) \Rightarrow F_a = mg\cos(30°) - N_1 \\ y': N_2 = mg\,\text{sen}(30°) \Rightarrow N_2 = \frac{1}{2}mg \Rightarrow N_2 = 5N \end{cases}$$

$\sum M_i = 0$ POLO: PUNTO DI APP. $\vec{N}_2$

$\Rightarrow M_{\vec{P}} = mg \cdot \frac{\ell}{2}$

$M_{\vec{N}_{1y}} = N_1 \cdot \frac{\sqrt{3}}{2} \cdot \ell$

$\Rightarrow \frac{1}{2}mg = \frac{\sqrt{3}}{2}N_1$

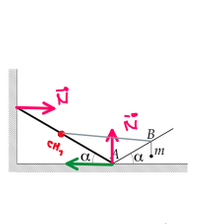
$\Rightarrow N_1 = \frac{10}{\sqrt{3}}$ N

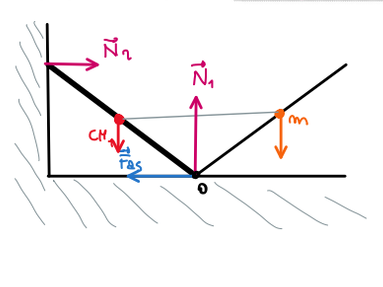
$\Rightarrow F_a = -\frac{10}{\sqrt{3}} + mg(\cos 30°) \simeq 2,89$ N

$F_{a\,max} = \frac{1}{\sqrt{2}} \cdot 5 \simeq 3,5$ N

$\Rightarrow F_a \le F_{a\,max} \Rightarrow$ EQUILIBRIO $\Rightarrow N_1 \simeq 5,77$ N

2. Un corpo rigido, formato da due aste saldate nel punto A, giace su di un piano orizzontale scabro avente coefficiente di attrito statico $\mu = 1/\sqrt{3}$. Una delle aste, lunga L e avente massa M, poggia su un muro verticale liscio. Sulla seconda asta, di massa trascurabile, è appesa una pallina di massa m. Sapendo che entrambe le aste formano un angolo $\alpha = 30°$ con l'orizzontale e che $M/m = 5$, determinare l'intervallo dei valori di $\overline{AB}$ per i quali il sistema è in equilibrio statico.





$\mu = \frac{1}{\sqrt{3}}$ MASSA ASTA: M = 5m

MASSA PUNTO: m

α = 30°

$\Rightarrow \vec{R} = \vec{P}_1 + \vec{P}_2 + \vec{F}_{as} + \vec{N}_1 + \vec{N}_2$

$$\Rightarrow \begin{cases} x: N_2 = F_a \\ y: N_1 = 6mg \end{cases}$$

$\sum M_i = 0$

$\Rightarrow$ POLO: O

$\Rightarrow M_{\vec{N}_2} = -N_2\,\text{sen}(30°) \cdot \ell$ $mg\,L\cos(30°)\left(\frac{5}{2} - x\right) - N_1\,\text{sen}(30°)$

$M_{\vec{P}_1} = 5mg \cdot \frac{1}{2}L\cos(30°)$

$M_{\vec{P}_2} = -mg \cdot x\,L\cos(30°)$

$\Rightarrow N_2\,\text{sen}(30°) = mg\cos(30°)\left(\frac{5}{2} - x\right)$

$\Rightarrow$ NEL CASO LIMITE IN CUI m È PIÙ A DESTRA POSSIBILE $\Rightarrow N_2 = 0$

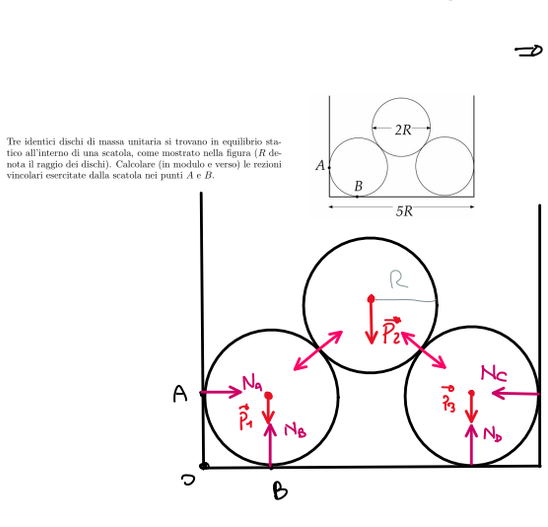
$\Rightarrow x = \frac{5}{2} \Rightarrow \overline{AB} = \frac{5}{2}L$

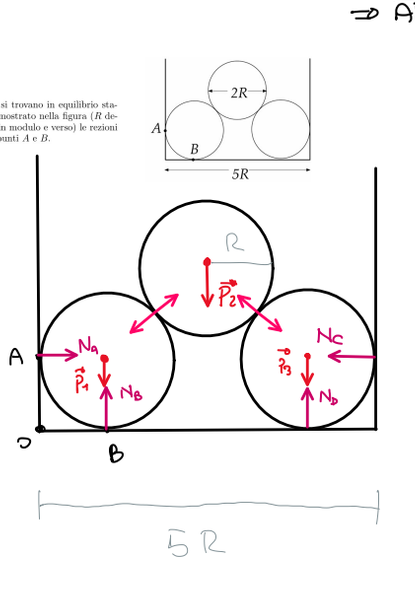
$\Rightarrow$ NEL CASO LIMITE IN CUI m È PIÙ A SINISTRA POSSIBILE $N_2 = F_a = F_{a\,max} = 2\sqrt{3}\,mg$

$\Rightarrow \frac{1}{\cancel{2}}2\sqrt{3}\,\cancel{mg} = \cancel{mg}\cos(30°)\left(\frac{5}{2} - x\right)$

$\Rightarrow \frac{5}{2} - x = \frac{\cancel{\sqrt{3}}}{\cancel{\sqrt{3}}} \cdot 2 \Rightarrow x = \frac{5}{2} - 2 = \frac{1}{2}$

$\Rightarrow \overline{AB} = \frac{1}{2}L$





$P = 3\vec{P}_1$

$\vec{R} = \vec{P}_1 + \vec{P}_2 + \vec{P}_3 + \vec{N}_A + \vec{N}_B + \vec{N}_C$

$$\Rightarrow \begin{cases} N_B + N_D = 3mg \Rightarrow 2N_B = 3mg \Rightarrow N_B = \frac{3}{2}mg \\ N_A = N_C \end{cases}$$

$\sum \vec{M}_i = 0$ POLO: O

$\Rightarrow M_{\vec{P}_1} = -mg \cdot R$

$M_{\vec{P}_2} = -mg \cdot \frac{5}{2}R$

$M_{\vec{P}_3} = -mg \cdot 4R$

$M_{N_C} = N_C \cdot R = N_A \cdot R$

$M_{N_D} = N_D \cdot 4R = N_B \cdot 4R = 6mg$

$\Rightarrow \left(1 + \frac{5}{2} + 4 - 6\right)mg = N_A \Rightarrow N_A = \frac{3}{2}mg = N_B$

$\Rightarrow \vec{N}_A = \frac{3}{2}mg \cdot \hat{x} = \frac{1}{2}Mg\,\hat{x}$

$\vec{N}_B = \frac{3}{2}mg \cdot \hat{y} = \frac{1}{2}Mg\,\hat{y}$